

Bifurcation of the Kuramoto model on networks

Tohoku University, AIMR

Hayato Chiba (千葉 逸人)

hchiba@tohoku.ac.jp

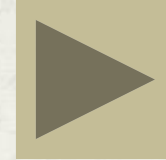
Oct 13, 2023

Various Synchronization

- Firefly flashing
- Neural network
- Cardiac muscle cell
- Network of power plants
- Locomotion



Firefly



Metronome



Pendulum



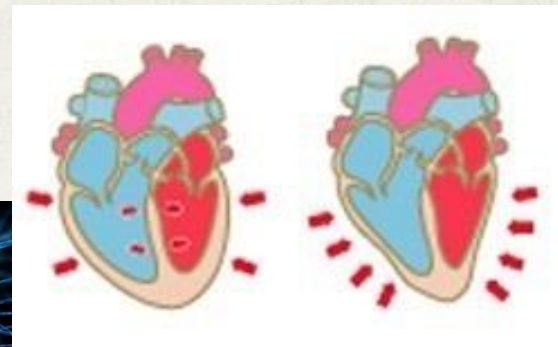
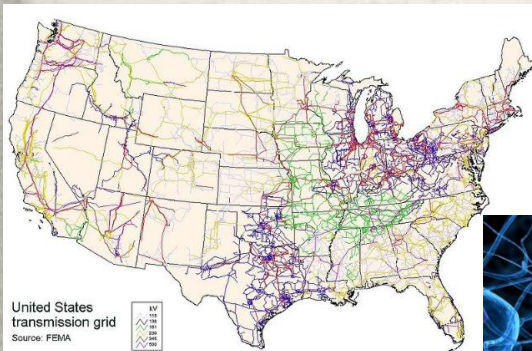
Bottoles



Bridge



Cat



Kuramoto model (1975)

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

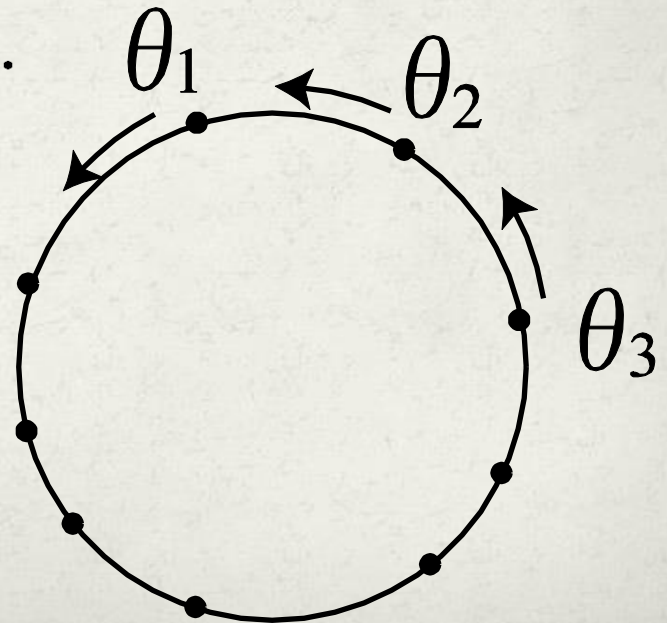
$$\theta_i \in S^1$$

N : the number of particles.

ω_i : natural frequency of θ_i .

$K > 0$: coupling strength.

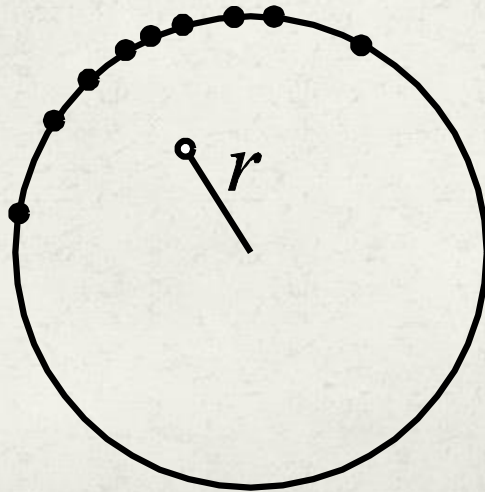
If $K > K_c$,
synchronization occurs.



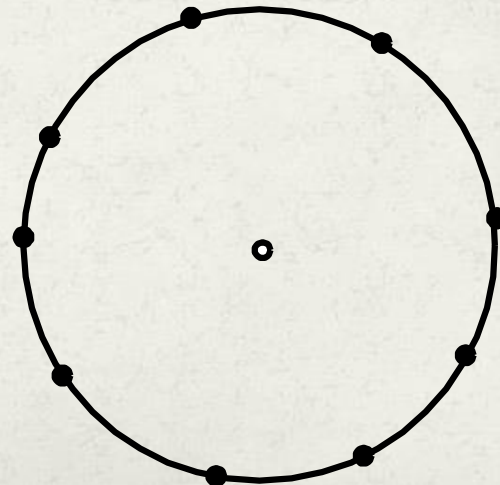
Kuramoto model (1975)

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$

Order parameter : $re^{i\psi} := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}$



synchro



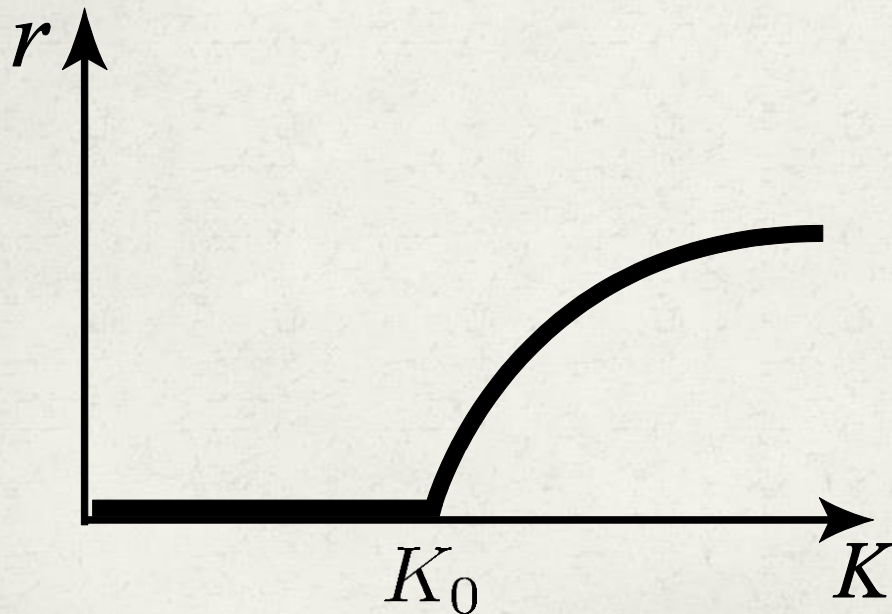
de-synchro

Kuramoto conjecture (1984)

$$N \rightarrow \infty,$$

$g(\omega)$: distribution function for ω_i 's.

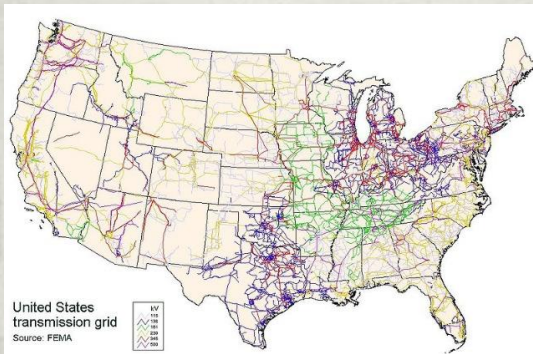
If $g(\omega)$ is even and unimodal, then



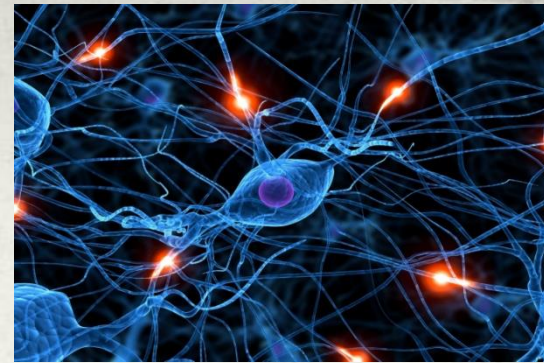
$$K_0 := \frac{2}{\pi g(0)}$$

H.C.,

A proof of the Kuramoto conjecture of the infinite dimensional Kuramoto model,
Ergo. Theo. Dyn. Syst, (2015).



Power grid



Neural Network

*Coupled oscillators
on
Large Networks*



muscle cell

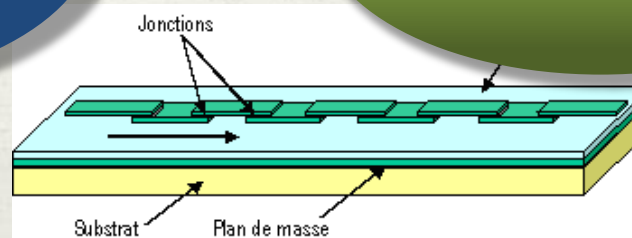


**Network
(discrete)**

Locomotion

VS

**Dynamics
(continuous)**



Josephson junction array

The Kuramoto model on a network:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N a_{ij} \sin(\theta_j - \theta_i).$$

A weighed graph with the adjacency matrix $A = \{a_{ij}\}_{i,j}^N$ is associated.

The theory of graph limits (by Lovasz et al, 2012).

$$A = \{a_{ij}\}_{i,j}^N \xrightarrow{\text{limit}} W(x, y) \quad (\text{graphon})$$

$W(x, y)$: a function on $I \times I$ ($I = [0, 1]$).

A node is labeled by a real number on $[0, 1]$.

$W(x, y)$ is a weight of the edge connecting x and y .

Assume that $W(x, y)$ is symmetric and square integrable.
 Define the Fredholm integral op. on $L^2(I, dx)$ by

$$(\hat{W}\phi)(x) = \int_I W(x, y)\phi(y)dy$$

ζ_M : the largest eigenvalue of $\hat{W}$
 (assumed to be positive) .

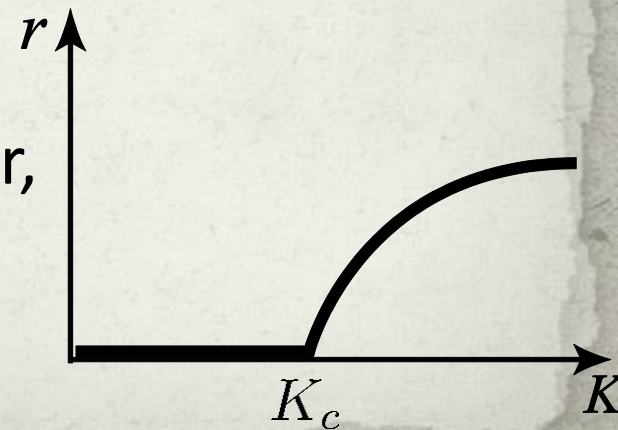
V_M : its eigenfunction.

Result.

Bifurcation value: $K_c = \frac{1}{\zeta_M} K_0$

The phase θ of a synchronized oscillator,
 labeled by x , is approximately given by

$$\theta \sim \arg(V_M(x)).$$



The continuous limit (Vlasov Eq.)

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta}(\rho V) = 0, \quad \rho = \rho(t, \theta, \omega, x),$$

The local order parameter (limit of $\frac{1}{N} \sum_{j=1}^N a_{ij} e^{i\theta_i}$)

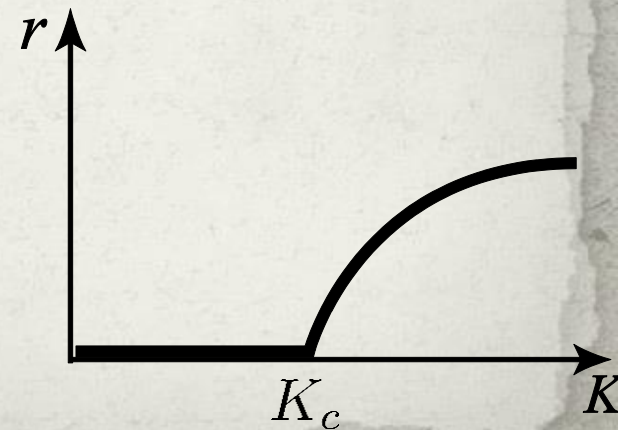
$$h(t, x) = \int \int \int W(x, y) e^{i\theta} \rho(t, \theta, \omega, y) g(\omega) d\theta d\omega dy$$

For $K > K_c = K_0/\zeta_M$, there is a stable steady state s.t.

$$h(t, x) = \text{const} \cdot V_M(x) \sqrt{K - K_c} + O(K - K_c)$$

Distribution of the phase locked osci.

$$\theta \sim \arg(V_M(x)).$$



Q. How does a structure of a graph $A = \{a_{ij}\}_{i,j}^N$ affect the dynamics (for large N)?

A. The eigenvalue-problem of the operator

$$(\hat{W}\phi)(x) = \int_I W(x, y)\phi(y)dy$$

determines a bifurcation structure.

Bifurcation value: $K_c = \frac{1}{\zeta_M} K_0$

The phase θ of a synchronized oscillator, labeled by x , is approximately given by

$$\theta \sim \arg(V_M(x)).$$

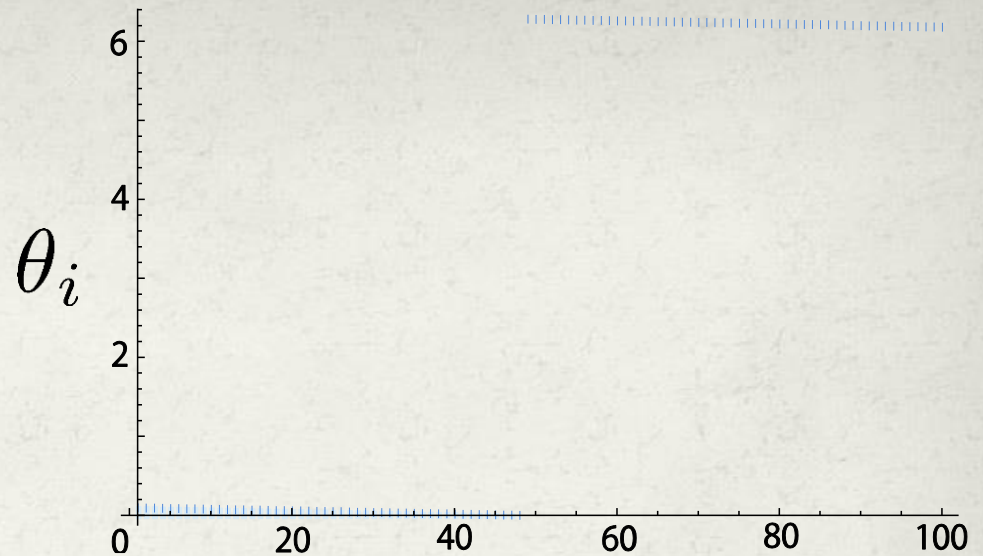
Numerical results.

For the classical (all-to-all)
Kuramoto model,

$$W(x, y) = 1,$$

$$\zeta_M = V_M(x) = 1.$$

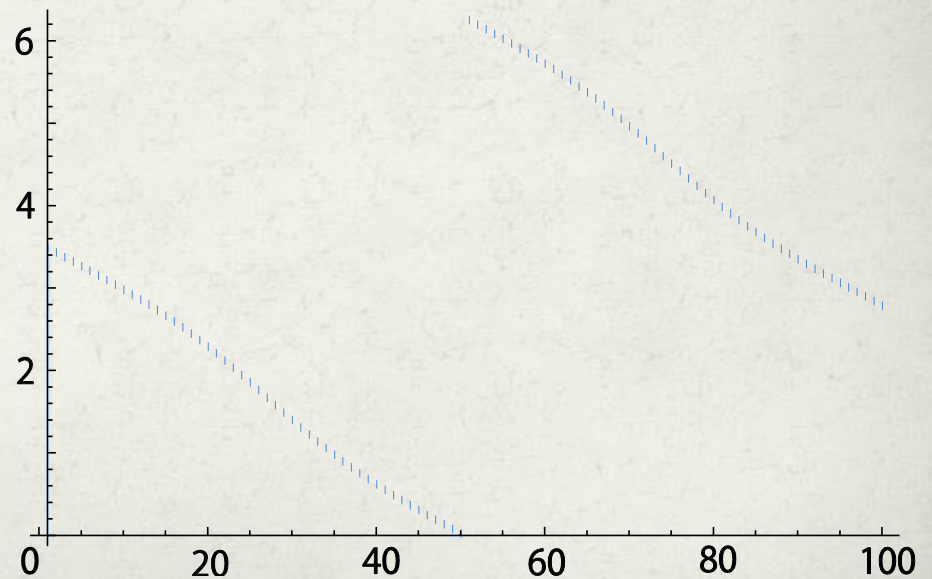
➡ $\theta \sim 0$



Put $W(x, y) = \cos 2\pi(x - y)$

Then, $V_M(x) = e^{\pm 2\pi i x}$

➡ $\theta \sim \pm 2\pi x.$



The label of oscillators

Design a network (two ways).

- Deterministic Weighted Graph

1. Find $W(x, y)$ s.t. the eigenfunction $V_M(x)$ takes a desired form.
2. Define a finite graph by a discretization.

$$a_{ij} = W(x_i, x_j), \quad x_j = j/N.$$

- Random 0-1 Graph

1. Find $W(x, y)$ s.t. the eigenfunction $V_M(x)$ takes a desired form.
2. Define a random graph by

$$\mathbb{P}(a_{ij} = 1) = W(x_i, x_j)$$

$$\mathbb{P}(a_{ij} = 0) = 1 - W(x_i, x_j)$$

Critical coupling strength

Dense graphs

- Complete $K_0 := \frac{2}{\pi g(0)}$
- Erdos-Renyi $K_c = K_0/p > K_0$
- Small-World $K_c = K_0/(2r + p - 4pr) > K_0$

Sparse graph

- Scale-Free
(power law) $W(x, y) = (1 - 2\gamma)(xy)^{-\gamma} \quad (0 < \gamma < 1/2)$
$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N^{1-\beta}} \sum_{j=1}^N a_{ij} \sin(\theta_j - \theta_i), \quad (\gamma < \beta < 1).$$

 $K_c < K_0.$
- D-dim lattice : Unknown.
(several authors observe sync. occurs only for $D \geq 3$.)

H.C, Medvedev, Mizuhara, Bifurcations in the Kuramoto model on graphs (2018).

Idea of the proof
~ A generalized spectral theory ~

Infinite- dimensional Kuramoto model

$$(KM_N) \begin{cases} \dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) = \omega_i + Kr \sin(\psi - \theta_i), \\ re^{i\psi} := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)}. \end{cases}$$



continuous limit

$$(KM_\infty) \begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta} (v \rho) = 0, \\ v := \omega + Kr \sin(\psi - \theta), \\ re^{i\psi} = \int_{\mathbf{R}} g(\omega) d\omega \int_0^{2\pi} e^{i\theta} \rho(\theta, \omega, t) d\theta. \end{cases}$$

Evolution eq. for $\rho = \rho(\theta, \omega, t)$: prob. dens. func. of $\theta \in S^1$ for each $(\omega, t) \in \mathbf{R} \times \mathbf{R}_+$.

Fourier coefficients:

$$Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta \in L^2(\mathbf{R}, g(\omega) d\omega).$$

Evolution equations on $\prod_{-\infty}^{\infty} L^2(\mathbf{R}, g(\omega) d\omega)$.

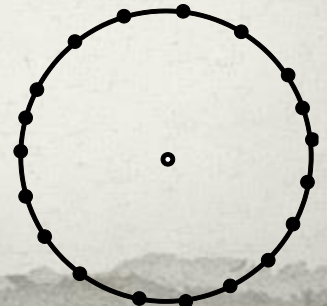
$$\begin{cases} \dot{Z}_1 = i\mathcal{M}Z_1 + \frac{K}{2}\mathcal{P}Z_1 + (\text{nonlinear}), \\ \dot{Z}_k = ik\mathcal{M}Z_k + (\text{nonlinear}), \quad (k = 2, 3, \dots), \\ re^{i\psi} := \mathcal{P}Z_1. \end{cases}$$

$\mathcal{M} : f(\omega) \mapsto \omega f(\omega)$, multiplication operator

$\mathcal{P} : f(\omega) \mapsto \int_{-\infty}^{\infty} f(\omega)g(\omega)d\omega$, projection

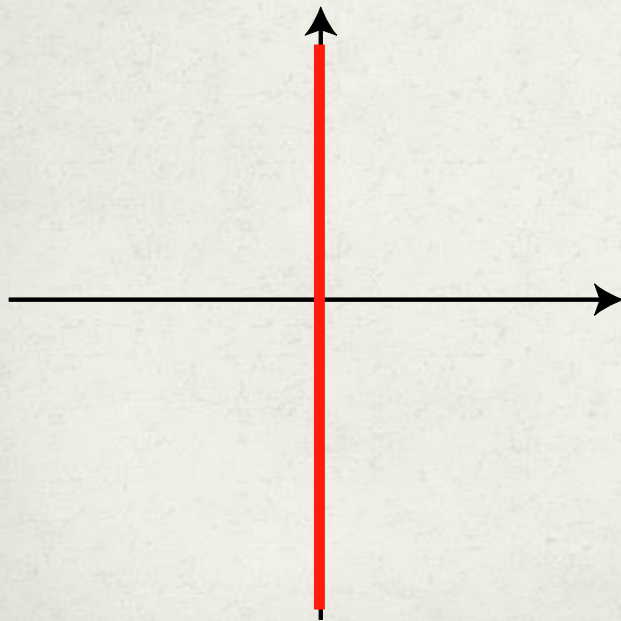
Trivial solution (de-synchro state)

$$Z_k \equiv 0 \iff \rho(\theta, \omega, t) = 1/(2\pi) \iff$$

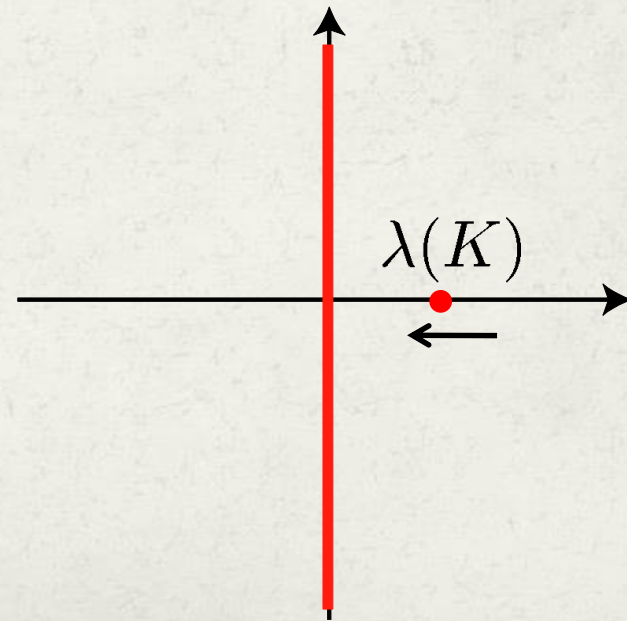


Spectrum of $T := i\mathcal{M} + K\mathcal{P}/2$

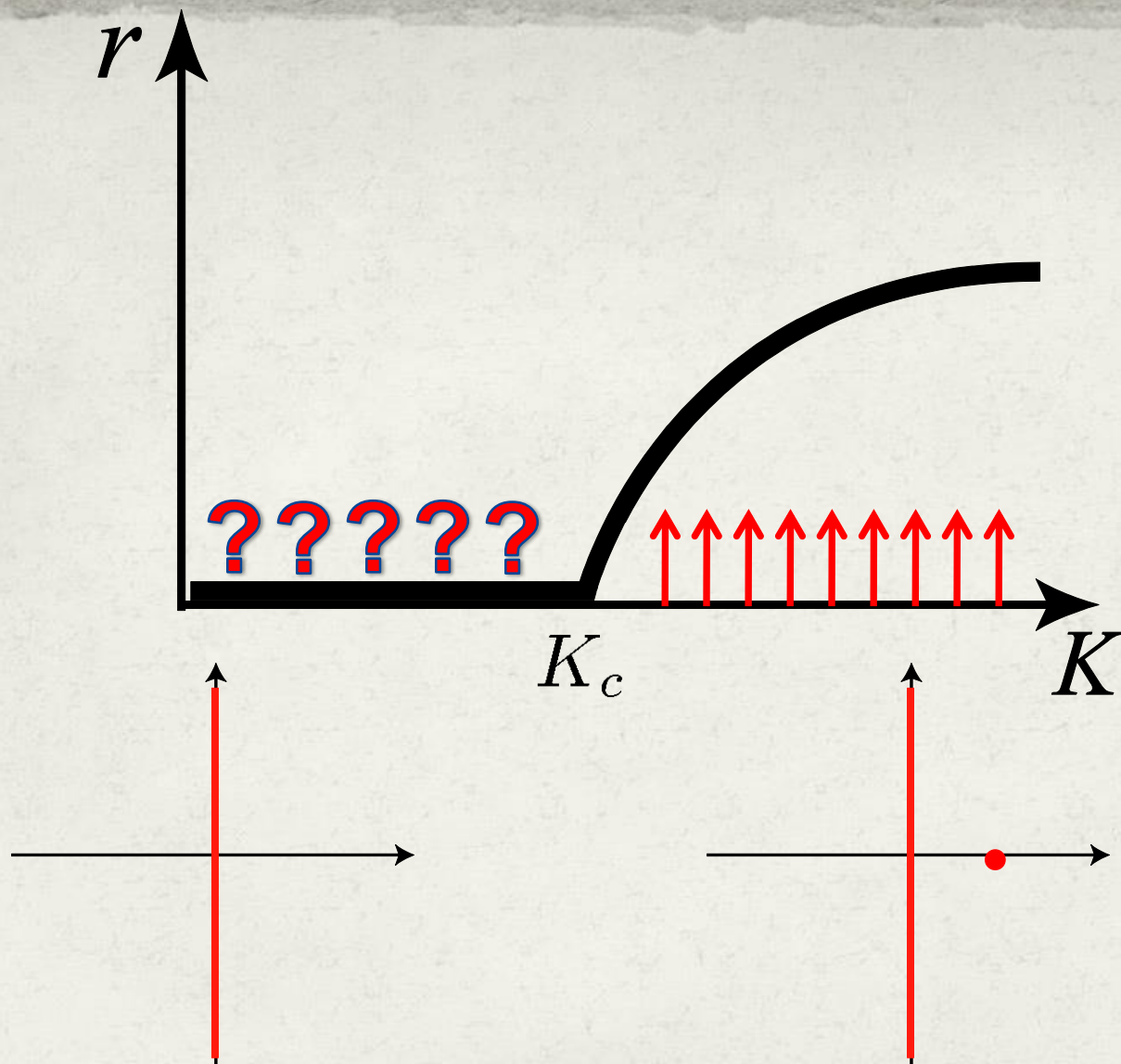
- conti. spec: $\sigma_c(T) = \sigma(i\mathcal{M}) = i\mathbf{R}$.
- eigenvalues: roots of $\int_{-\infty}^{\infty} \frac{1}{\lambda - i\omega} g(\omega) d\omega = 2/K$,
which exist iff $K_c := 2/(\pi g(0)) < K$.



$$0 < K < K_c$$



$$K_c < K$$



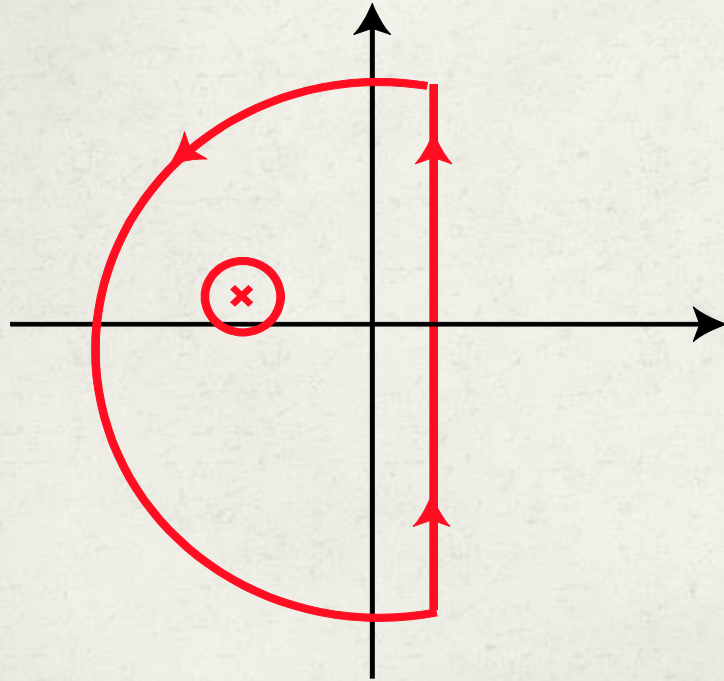
Stability? Bifurcations?

Let's reformulate in terms of the Gelfand triplet.

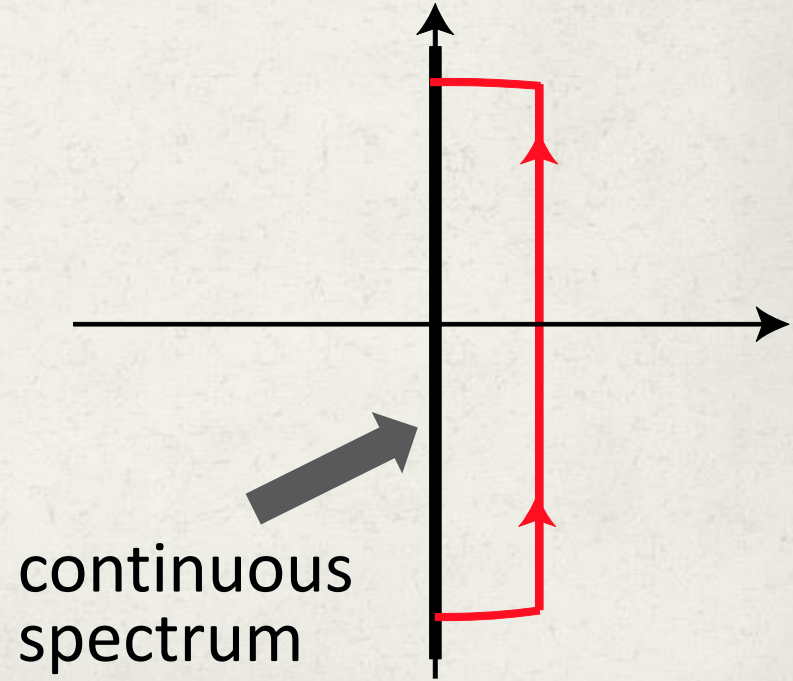
$$\dot{x} = \mathcal{A}x, \quad \mathcal{A} : X \rightarrow X \quad \text{Laplace inversion.}$$

$$e^{\mathcal{A}t}\phi = \frac{1}{2\pi i} \int_{x-i\infty}^{x+i\infty} e^{\lambda t} (\lambda - \mathcal{A})^{-1} \phi d\lambda$$

Resolvent. $R_\lambda \phi = (\lambda - \mathcal{A})^{-1} \phi$, analytic w.r.t. λ on $\mathbb{C} \setminus \sigma(\mathcal{A})$



$$e^{\mathcal{A}t}\phi = \sum (\text{residues})$$



$$e^{\mathcal{A}t}\phi = ?$$

Eq. $\dot{x} = \mathcal{A}x$, $\mathcal{A} : X \rightarrow X$

Sol. $e^{\mathcal{A}t} \phi = \frac{1}{2\pi i} \int_{x-i\infty}^{x+i\infty} e^{\lambda t} (\lambda - \mathcal{A})^{-1} \phi d\lambda$

Resolvent. $R_\lambda \phi = (\lambda - \mathcal{A})^{-1} \phi$, analytic w.r.t. λ on $\mathbf{C} \setminus \sigma(\mathcal{A})$

The resolvent diverges on $\lambda \in \sigma(\mathcal{A})$ as a function in X ...
But it may converge in a larger space.

e.g. Gaussian.

$$f(x) = \frac{1}{\sqrt{2\pi s}} \exp \left[-\frac{x^2}{2s^2} \right] \xrightarrow{s \rightarrow +0} \begin{cases} \infty & \text{(as a usual func.)} \\ \delta & \text{(Dirac delta.)} \end{cases}$$

Idea: Consider the multiplication operator on $L^2(\mathbf{R})$.

$$x : \phi(x) \mapsto x\phi(x).$$

Resolvent. $R_\lambda \phi = (\lambda - x)^{-1} \phi(x)$.

Spectrum. $\sigma(\mathcal{A}) = \mathbf{R}$.

When $\lambda \in \mathbf{R}$, $R_\lambda \phi \notin L^2(\mathbf{R})$.

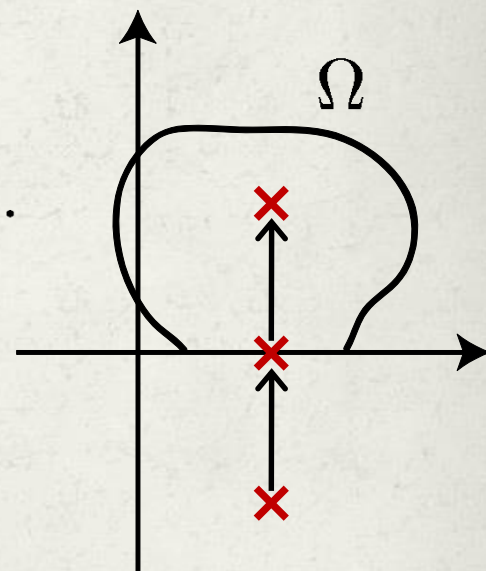
However, the inner product may exist!

$$(R_\lambda \phi, \psi) = \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx, \quad (\text{Im}(\lambda) < 0).$$

$$\downarrow \text{ (if } \phi \text{ and } \psi \text{ are continuous)}$$
$$\lim_{\text{Im}(\lambda) \rightarrow -0} \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx$$

$$\downarrow \text{ (if } \phi \text{ and } \psi \text{ are holomorphic)}$$

$$\int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx + 2\pi i \phi(\lambda) \psi(\lambda), \quad (\text{Im}(\lambda) > 0).$$



The analytic continuation

$$(R_\lambda \phi, \psi) = \int_{\mathbf{R}} \frac{1}{\lambda - x} \phi(x) \psi(x) dx + 2\pi i \phi(\lambda) \psi(\lambda), \quad (\text{Im}(\lambda) > 0).$$

exists only when ϕ and ψ are holomorphic.

$X \subset L^2(\mathbf{R})$: some class of holomorphic functions.

X' : the dual space (the set of conti. linear functionals on X)

The mapping $\psi \mapsto (R_\lambda \phi, \psi)$ defines a linear functional on X .

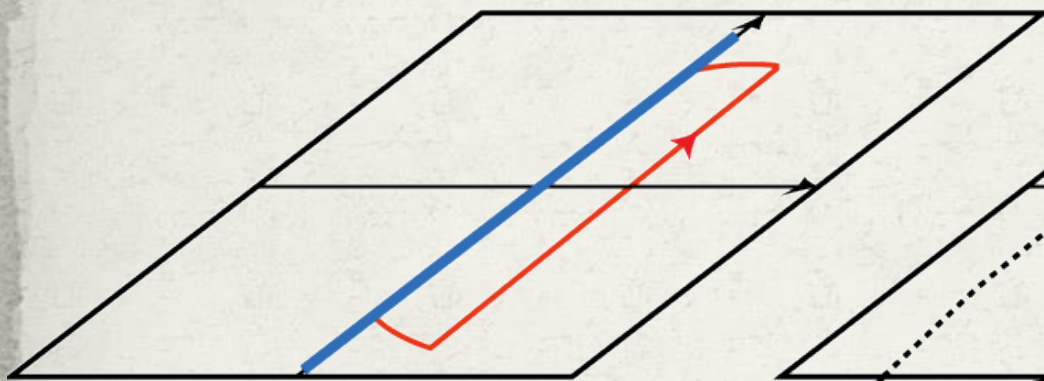
The mapping $\phi \mapsto (R_\lambda \phi, \cdot) \in X'$ defines a linear map from X into X' .

The resolvent $R_\lambda : L^2(\mathbf{R}) \rightarrow L^2(\mathbf{R})$ has an analytic continuation from the lower half plane to the upper half plane as an operator from X into X' .

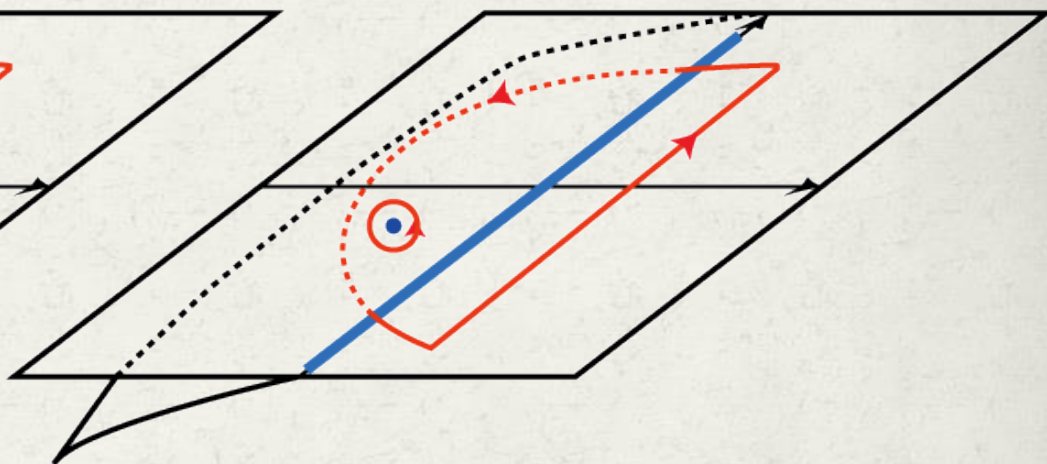
Gelfand triplet. $X \subset L^2(\mathbf{R}) \subset X'$

spectrum $\longleftrightarrow$ singularities of the resolvent (on $\mathbb{C}$).
generalized spectrum $\longleftrightarrow$ singularities of an analytic continuation of the resolvent (on a Riemann surface).

The first Riemann sheet



The second Riemann sheet



Through the Laplace inversion formula, the generalized spectrum induces an exponential decay of a solution.

- H. Chiba, **Adv. in Math.** 273, 324-379, (2015).

A spectral theory on a Gelfand triplet.

For the Kuramoto model,

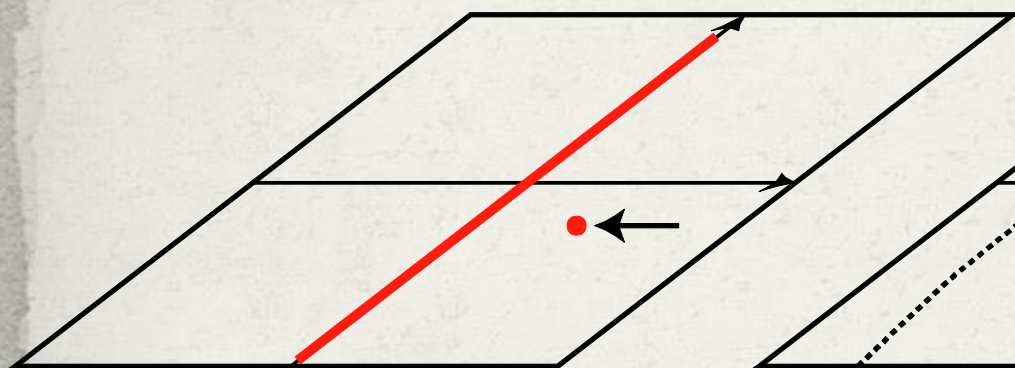
$$\left\{ \begin{array}{l} T = i\mathcal{M} + \frac{K}{2}\mathcal{P}. \\ X : \text{a certain space of holo. func.} \\ \text{Gelfand triplet : } X \subset L^2(\mathbf{R}, g(\omega)d\omega) \subset X' \\ \text{paring : } \langle \mu | \phi \rangle \\ g(\omega) : \text{Gaussian distribution.} \end{array} \right.$$

The resolvent $(\lambda - T)^{-1} : L^2 \rightarrow L^2$ has an analytic continuation $\mathcal{R}_\lambda : X \rightarrow X'$.

- generalized eigenvalues λ_n
- generalized eigenfunctions $\mu_n \in X'$
- generalized eigenspaces and projections.

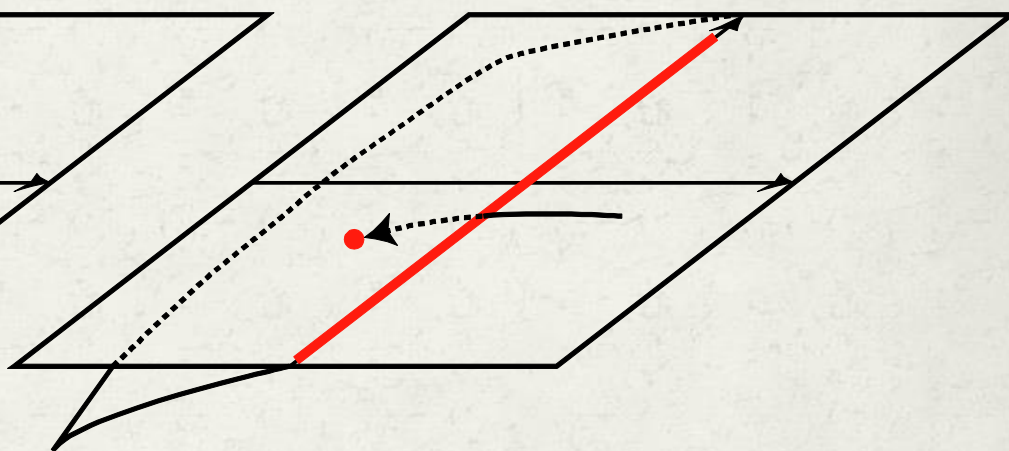
For the Kuramoto model, the number of generalized eigenvalues is countable.
They lie on the **second** Riemann sheet of the resolvent.

Spectrum in L^2 -sense



$$K_c < K$$

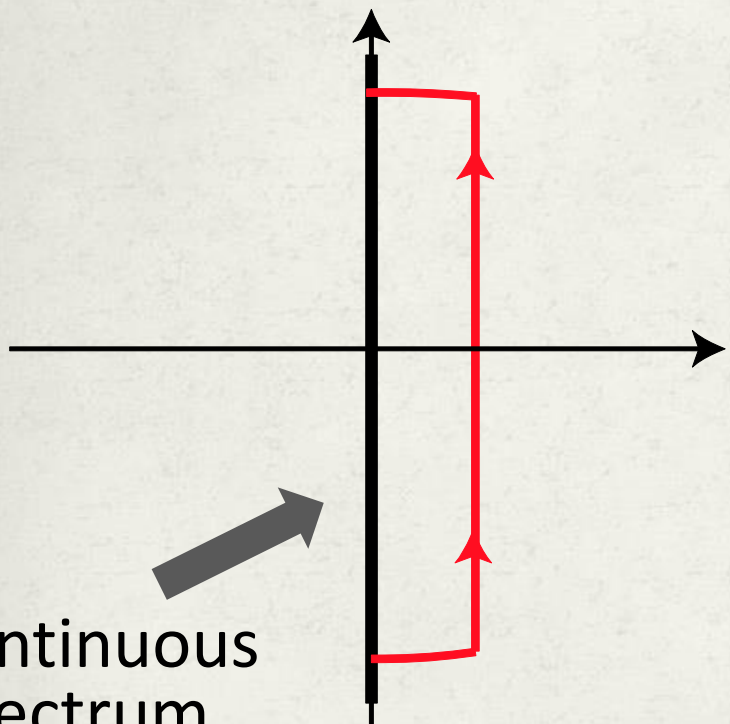
Generalized spectrum



$$0 < K < K_c$$

Semigroup in L^2 -sense.

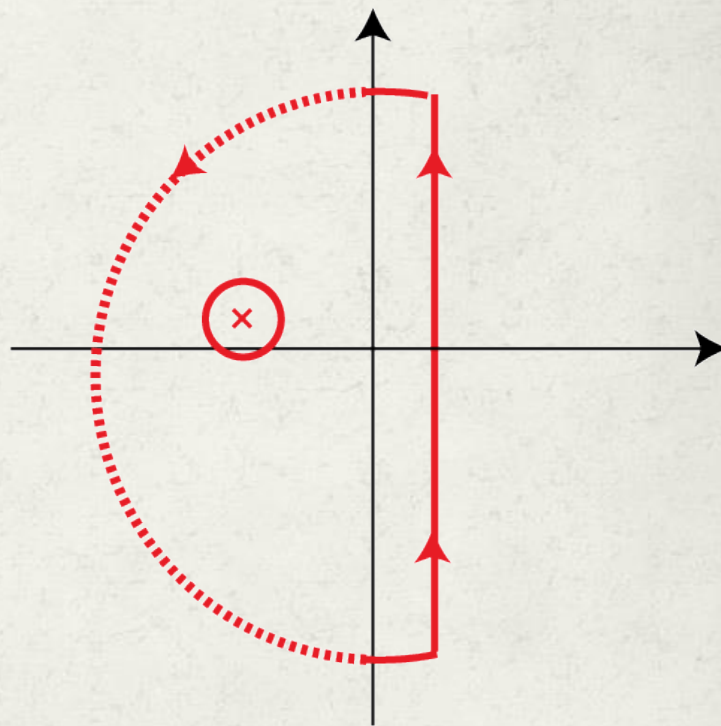
$$e^{Tt}\phi = \frac{1}{2\pi i} \int_{\gamma} e^{\lambda t} (\lambda - T)^{-1} \phi d\lambda$$



continuous
spectrum

Semigroup on the dual sp.

$$(e^{Tt})^{\times} \phi = \frac{1}{2\pi i} \int_{\gamma} e^{\lambda t} \cdot \mathcal{R}_{\lambda} \phi d\lambda$$



$$(e^{Tt})^{\times} \phi = \sum (\text{residues})$$

Theorem. (spectral decomposition)

For any $\phi \in X$,

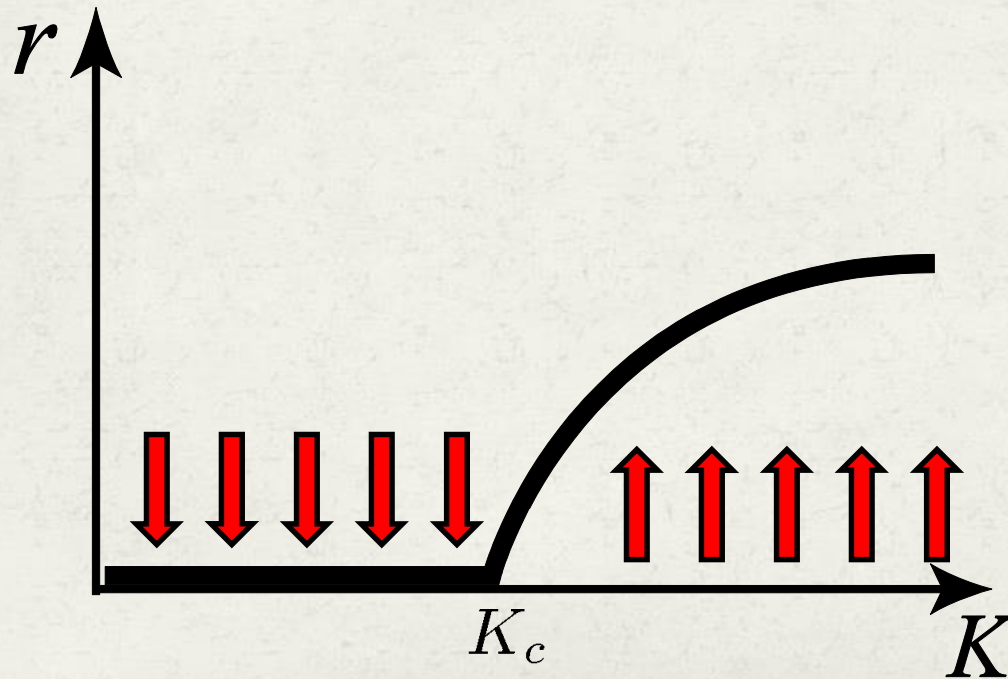
$$(e^{Tt})^\times \phi = \sum_{n=0}^{\infty} e^{\lambda_n t} \langle \mu_n | \phi \rangle \cdot \mu_n.$$

Theorem. (completeness)

- (i) A system of generalized eigenfunctions $\{\mu_k\}_{k=0}^{\infty}$ is complete
($\langle \phi | \mu_k \rangle = 0, \forall k \Rightarrow \phi = 0$).
- (ii) linearly independent : $\sum a_k \mu_k = 0 \Rightarrow a_k = 0$
- (iii) the decomposition is unique.

$$(e^{Tt})^\times \phi = \sum_{n=0}^{\infty} e^{\lambda_n t} \langle \mu_n | \phi \rangle \cdot \mu_n.$$

When $0 < K < K_c$, all gener. eigenvalues lie on the left half plane, which proves the stability of the de-synchro state.



The next purpose is a bifurcation at $K = K_c$.

On $L^2(\mathbf{R}, g(\omega)d\omega)$, the center subspace is infinite-dim because $\sigma_c(T) = i\mathbf{R}$.

Generalized center subspace:

$$\mathbf{E}_c = \text{span}\{ \mu_n \mid \lambda_n \in i\mathbf{R} \} \subset X'.$$



Existence of a finite-dim center mfd. on X' .

For the Kuramoto model, this is one-dimensional.

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N.$$



$N \rightarrow \infty$

(continuous model)



$$Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta.$$

$$\frac{dZ_1}{dt} = TZ_1 - \frac{K}{2}(P_0, Z_1)Z_2.$$



$$X \subset L^2(\mathbf{R}, g(\omega)d\omega) \subset X' \quad \text{inclusion.}$$

$$\frac{d}{dt}|Z_1\rangle = T^\times|Z_1\rangle - \frac{K}{2}\langle P_0 | Z_1\rangle|Z_2\rangle.$$



$$|Z_1\rangle = \alpha(t)|\mu_0\rangle + |Y_1\rangle.$$

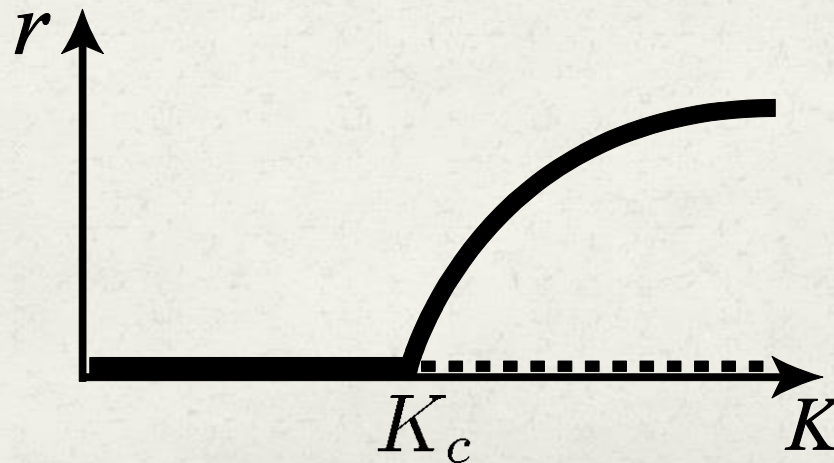
$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + O(|\alpha|^5).$$

Dynamics on a center manifold.

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + O(|\alpha|^5).$$

$$|\alpha| = \sqrt{\frac{-p_1}{p_3}} \sqrt{K - K_c}.$$

$$p_1 = D_0/K_c, \quad p_3 = \pi D_0 K_c^3 g''(0)/16.$$



Kuramoto conjecture is proved.

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N (\sin(\theta_j - \theta_i) + h \sin 2(\theta_j - \theta_i)).$$

 $N \rightarrow \infty$

(continuous model)

 $Z_k(t, \omega) := \int_0^{2\pi} e^{ik\theta} \rho(\theta, \omega, t) d\theta.$

$$\frac{dZ_1}{dt} = TZ_1 - \frac{K}{2}(P_0, Z_1)Z_2 - \frac{Kh}{2}(Z_2, P_0)Z_{-1}$$

 $i : X \rightarrow X'$

$$\frac{d}{dt}|\bar{Z}_1\rangle = T^\times|Z_1\rangle - \frac{K}{2}\langle P_0 | Z_1\rangle|Z_2\rangle - \frac{Kh}{2}\langle P_0 | Z_2\rangle|Z_{-1}\rangle.$$

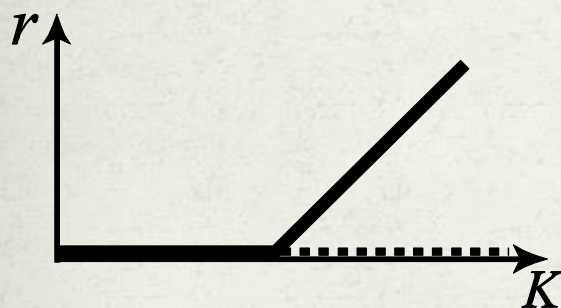
 $|Z_1\rangle = \alpha(t)|\mu_0\rangle + |Y_1\rangle.$

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_3\alpha|\alpha|^2 + p_2\alpha^2 e^{-i\arg(\alpha)} + O(|\alpha|^4).$$

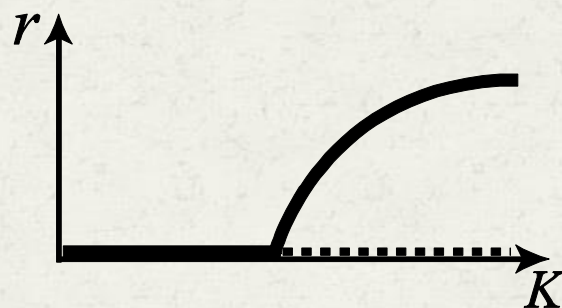
Dynamics on a center manifold.

$$\frac{d}{dt}\alpha = (K - K_c)p_1\alpha + p_2\alpha^2 e^{-i\arg(\alpha)} + O(|\alpha|^3).$$

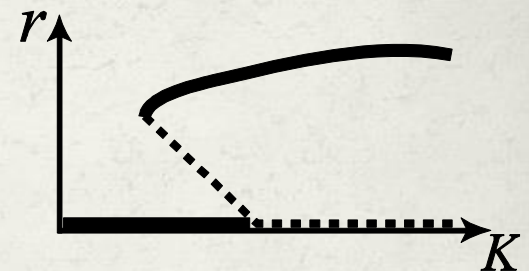
$$|\alpha| = \frac{2(1-h)}{K_c^3 \gamma h} (K - K_c).$$



$$h < 0$$



$$h = 0$$



$$0 < h < 1$$

- H.Chiba, **Ergo. Theo. Dyn. Syst**, (2013).
- H.Chiba, I.Nishikawa, **Chaos**, 21, 043103, (2011)

Applications of the generalized spectral theory

- The generalized spectral theory

H. C.

A spectral theory of linear operators on rigged Hilbert spaces under certain analyticity conditions. **Adv. in Math.** 273, 324-379, (2015).

- A proof of the Kuramoto conjecture.

H. C.

Ergo. Theo. Dyn. Syst, (2015).

- Applications to Schrödinger operators / reaction diffusion eqs.

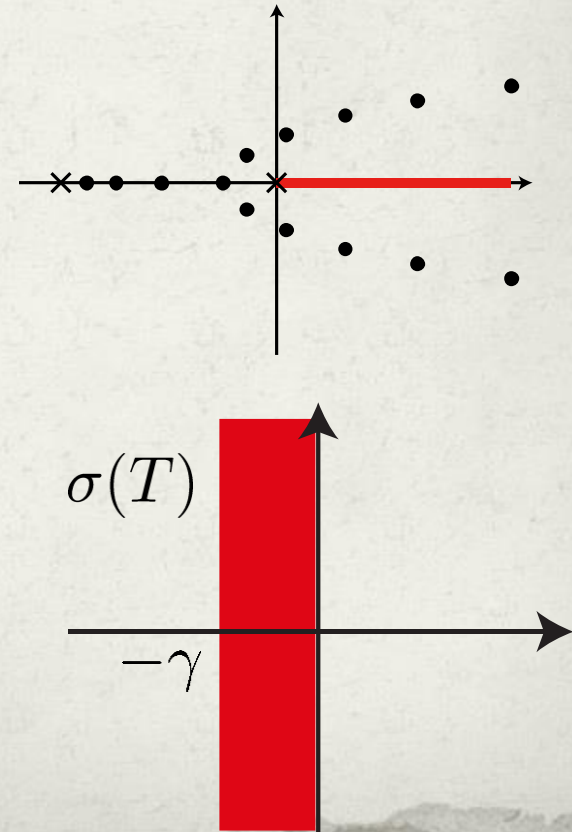
H.C.

Kyushu Journal of Math. (2018).

- The second order Kuramoto model
(with G. Medvedev, arXiv:2106.07784).

$$\ddot{\theta}_i + \gamma \dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N a_{ij} \sin(\theta_j - \theta_i)$$

$$K_c = \left(\mu_{max} \pi g(0) - \int \frac{g(\omega)}{1 + \omega^2} d\omega \right)^{-1}$$

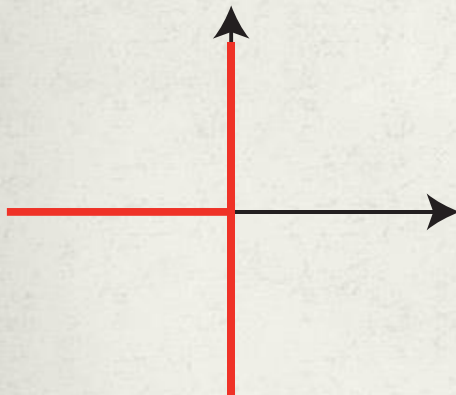


- Collective firing of neurons.
(with K.Kotani, **Physica D**, 2021)

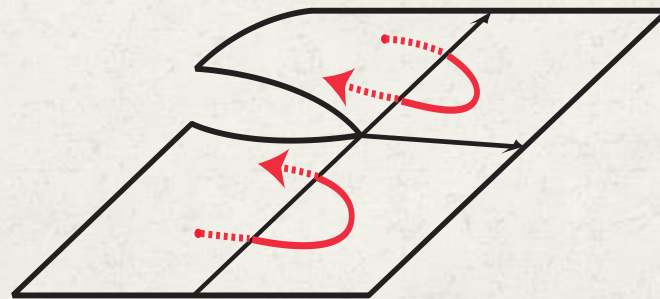


$$\frac{dV_i}{dt} = g_L \frac{(V_i - V_R)(V_i - V_T)}{V_T - V_R} + I_i - g_{syn}(t)(V_i - V_{syn})$$

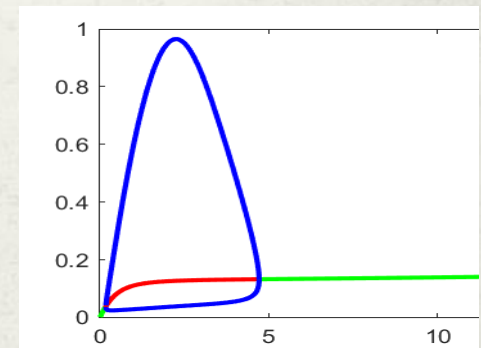
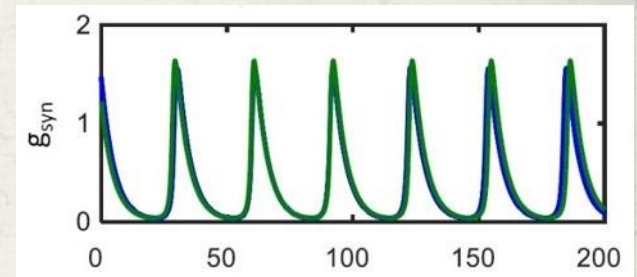
$$\frac{dg_{syn}}{dt} = -\frac{1}{\tau} g_{syn} + g_{peak} \sum \delta(t - t_k)$$



Continuous spectrum



Motion of the gener. e.values
on the Riemann surface.

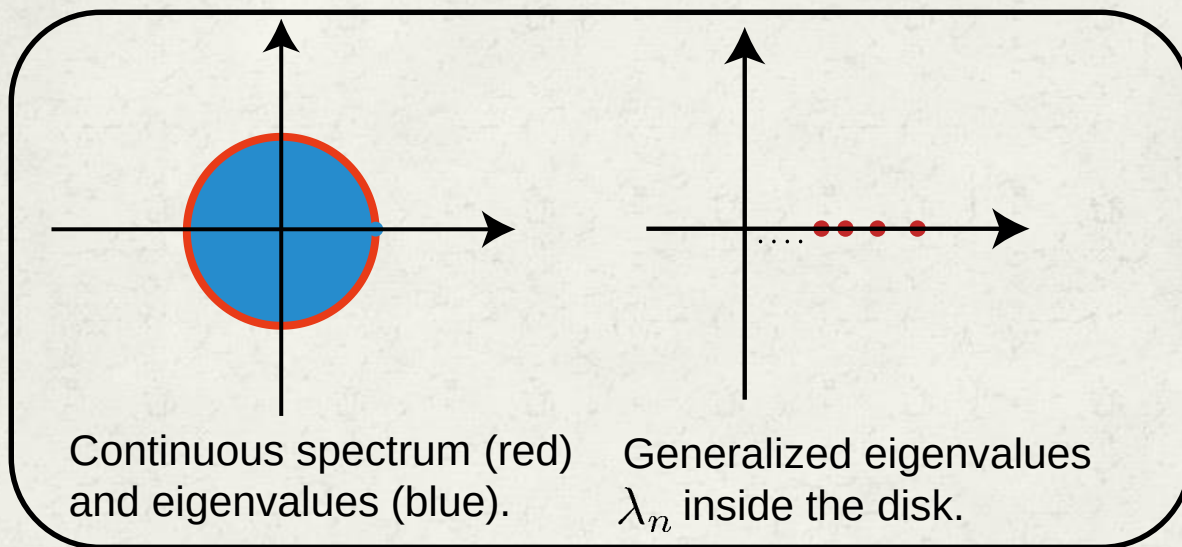


Hopf bifurcation of the
brain wave (gamma wave).

- Perron-Frobenius / Koopman op. of symbolic dynamical systems
(with I. Ishikawa, A. Ikeda, arXiv:2106.12087).

Left shift. $S : \omega = (\omega_1, \omega_2, \dots) \mapsto (\omega_2, \omega_3, \dots)$

Koopman. $T_S(\varphi)(\omega) := \varphi \circ (S\omega), \quad \varphi \in L^2(\Sigma, d\mu)$



Spectral decomposition theorem:

$$T_S^* \varphi = \sum_{n=0}^{\infty} \lambda_n \langle B'_n | \varphi \rangle B_n, \quad T_S^k \varphi \rightarrow \int_{\Sigma} \varphi d\mu. \quad (k \rightarrow \infty)$$

λ_n gives the rate of convergence to the mixing state.